

Assignment IIII

Machine Learning

February 14, 2026

Abstract

These problems aim to review the Decision Tree (DT) and Random Forest (RF) models for classification and regression scenarios. Please upload your solutions to Classroom before February 24th.

Problem 1

Describe the domain and image of the following functions:

- a) Gini impurity function.
- b) KL Divergence.

Problem 2

Consider a set $S = \{a, b, c, d, e, f, g\}$. In the context of bagging,

- a) What is the probability of getting a subset of only one element?
- b) What is the probability of getting the original set?

Problem 3

Consider the dataset from [1]. Make a Python script to solve the regression problem of forecasting city and highway fuel consumption with sklearn using the DT model. Write down all the assumptions, hyperparameters, and data preprocessing operations you make.

Problem 4

Consider the following function:

$$f(x) = 1 - 2^{\sin(-x^2)}, \quad x \in \mathcal{I} = [-\pi, \pi].$$

Make a Python script to solve the regression problem associated with a DT using sklearn. Write down all the assumptions you make. Also, write down the hyperparameters you choose and explain why you picked them that way.

Problem 5

Consider the dataset from [2]. Make a script using sklearn to solve the binary classification problem associated with classifying whether a given horse will make it or not. Use a DT model. Write down all the assumptions and data preprocessing operations you make. Incorporate the F1 score to explain your findings.

Problem 6

Consider the dataset from [3]. Make a script using sklearn to solve the multiclass classification problem associated with classifying beers by style. Use a DT model. Write down all the assumptions and data preprocessing operations you make. Then, incorporate a confusion matrix into your analysis.

Problem 7

Consider the general quadratic equation

$$ax^2 + bx + c = 0.$$

Do your best to classify the roots of the equation above in terms of its coefficients using a DT model with sklearn. For simplicity, you must fix the domain for x and propose a suitable domain for the coefficients. Write down all your assumptions and show off your findings using some plots.

Problems 8 to 12

Repeat problems 3 to 7 using the RF model from sklearn. Write down the hyperparameters you choose and all your assumptions.

References

- [1] <https://www.kaggle.com/datasets/rinichristy/2022-fuel-consumption-ratings>
- [2] <https://www.kaggle.com/datasets/yasserh/horsesurvivalprognostication>
- [3] <https://www.kaggle.com/datasets/ankurnapa/ankurs-beer-data-set>